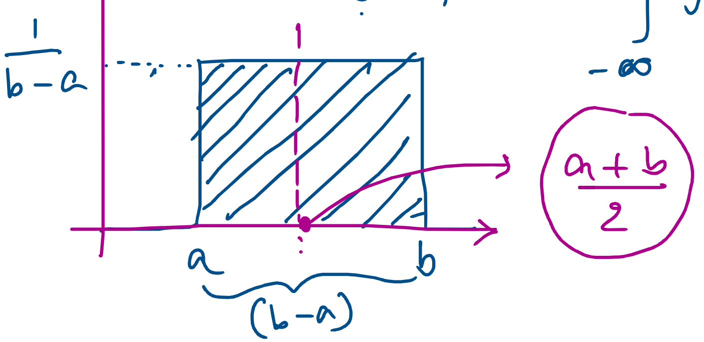
$$f(x) = \frac{1}{b-a} \; ; \; a \le x \le b$$
$$\quad = 0 \; ; \; \text{oth}$$



$$\int_{-\infty}^{+\infty} f(x)\,dx = \underbrace{\int_{-\infty}^{a} (0)\,dx}_{0} + \int_{a}^{b} \left(\frac{1}{b-a}\right) dx + \underbrace{\int_{b}^{\infty} (0)\,dx}_{0}$$

$$= \int_{a}^{b} \left(\frac{1}{b-a}\right) dx = \left(\frac{1}{b-a}\right) \int_{a}^{b} dx = \left(\frac{1}{b-a}\right) \cdot x\Big]_{a}^{b}$$

$$= \frac{1}{\cancel{(b-a)}} \cdot \cancel{(b-a)}$$

$$= 1$$

$$\int dx = \int 1 \cdot dx = x + c$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x \cdot f(x)\,dx = \int_{a}^{b} \frac{x}{b-a}\,dx = \frac{1}{b-a} \cdot \int_{a}^{b} x\,dx = \frac{1}{2(b-a)} \cdot \left[x^2\right]_{a}^{b}$$

$$= \frac{1}{2(b-a)} \cdot (b^2 - a^2)$$

$$= \frac{1}{2\cancel{(b-a)}} (b+a) \cancel{(b-a)}$$

$$= \frac{a+b}{2}$$

$$\mathrm{Var}[X] = \mathbb{E}[X^2] - \left(\mathbb{E}[X]\right)^2$$

Exponential Distribution:-

$$f(x) = \lambda e^{-\lambda x} \quad \Big| \; 0 \le x \le \infty$$

$$\int_{-\infty}^{\infty} f(x)\,dx = \int_{0}^{\infty} f(x)\,dx = \int_{0}^{\infty} \lambda e^{-\lambda x}\,dx$$

$$= \lambda \int_{0}^{\infty} e^{-\lambda x}\,dx$$

$$= \lambda\left(-\frac{1}{\lambda}\right) \cdot \left[e^{-\lambda x}\right]_{0}^{\infty}$$

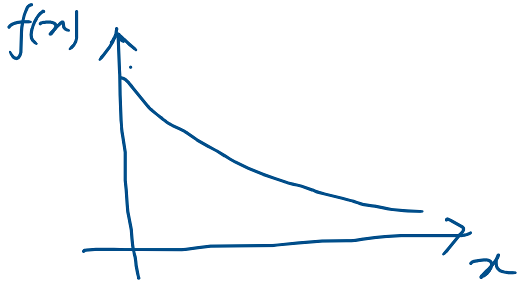
$$= (-1)\,(0-1) = 1$$

Poisson PMF

$$P(X=x) = \frac{e^{-\lambda}\lambda^{x}}{x!}$$

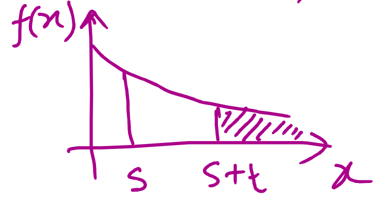
$x \in W$ $\quad W=\{0,1,2,\dots\}$

$$\int e^{ax}\,dx = \frac{e^{ax}}{a} + c$$



Bus-stop: you have waited for 10 mins.
What is the prob of the bus arriving after 5 minutes.
The arrival of the bus is random process (poisson)

λ

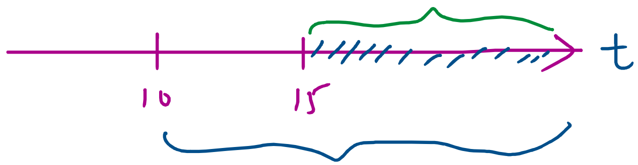


$$P(t > 15 \mid t \geq 10) = \frac{P((t>15) \cap (t \geq 10))}{P(t \geq 10)} = \frac{P(t>15)}{P(t \geq 10)}$$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

$$(t > 15) \cap (t \geq 10) = (t > 15)$$

$$P(t > 15 \mid t \geq 10) = P(t > 5)$$



$$P(x > s+t \mid x > s) = \frac{P(x > s+t)}{P(x > s)}$$

Now $P(x > s+t) = \int_{s+t}^{\infty} \lambda e^{-\lambda x} dx = \lambda \cdot \left[\frac{e^{-\lambda x}}{-\lambda}\right]_{s+t}^{\infty}$

$$= (-1)\left[0 - e^{-\lambda(s+t)}\right]$$

$$= e^{-\lambda(s+t)}$$

$$P(X>S) = \int_s^\infty \lambda e^{-\lambda x}\, dx = \cancel{\lambda} \cdot \left(-\frac{1}{\cancel{\lambda}}\right) \cdot \left[e^{-\lambda x}\right]_s^\infty$$

$$= (-1) \cdot \left(0 - e^{-\lambda s}\right) = e^{-\lambda s}$$

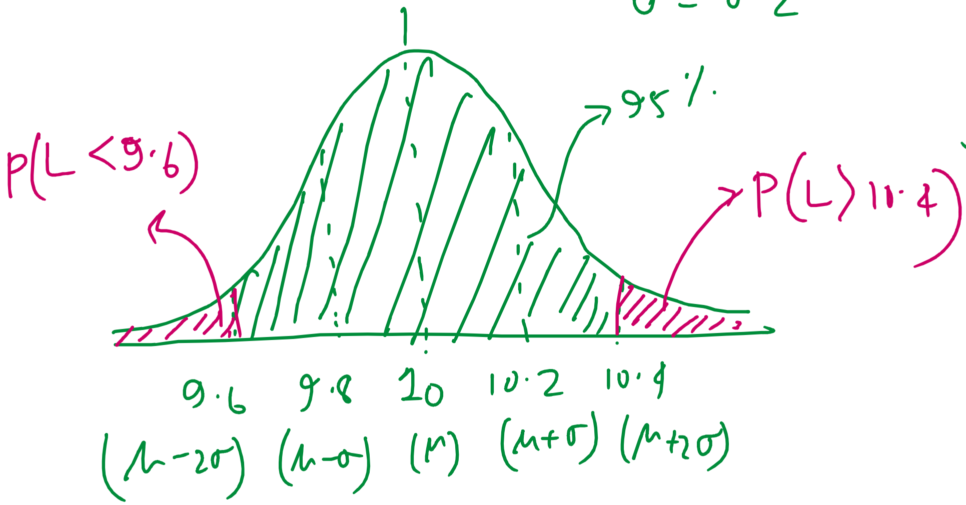
$$\therefore \quad P(X > s+t \mid X > s) = \frac{P(X > s+t)}{P(X > s)} = \frac{e^{-\lambda(s+t)}}{e^{-\lambda s}} = e^{-\lambda t}$$

$$P(X>t) = e^{-\lambda t}$$

$$\boxed{P(X > s+t \mid X > s) = P(X > t)}$$

Memoryless Property.

$\sigma = 0.2$



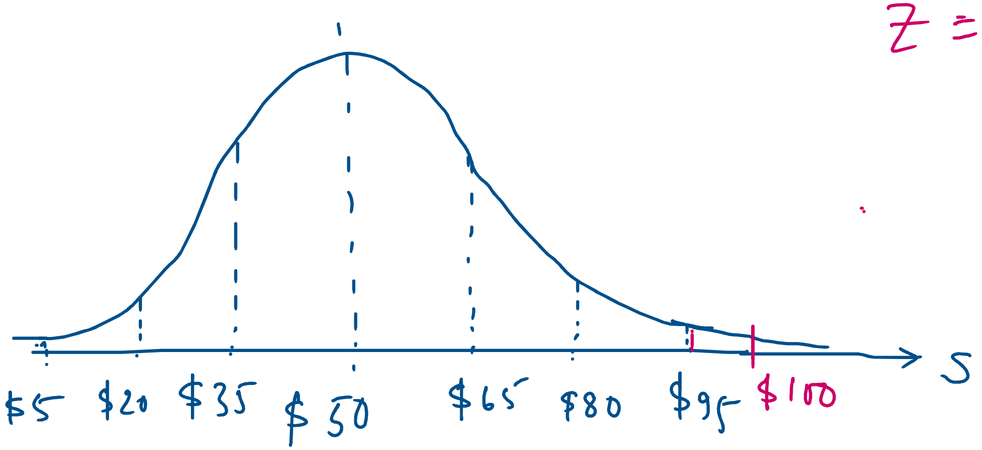
$$P(9.6 \le L \le 10.4) = 0.95$$

$$\therefore P(L<9.6) + P(L>10.4) = 1 - 0.95$$

$$= 0.05$$

5% defect.

$\mu = 50\$$ & $\sigma = \$15$



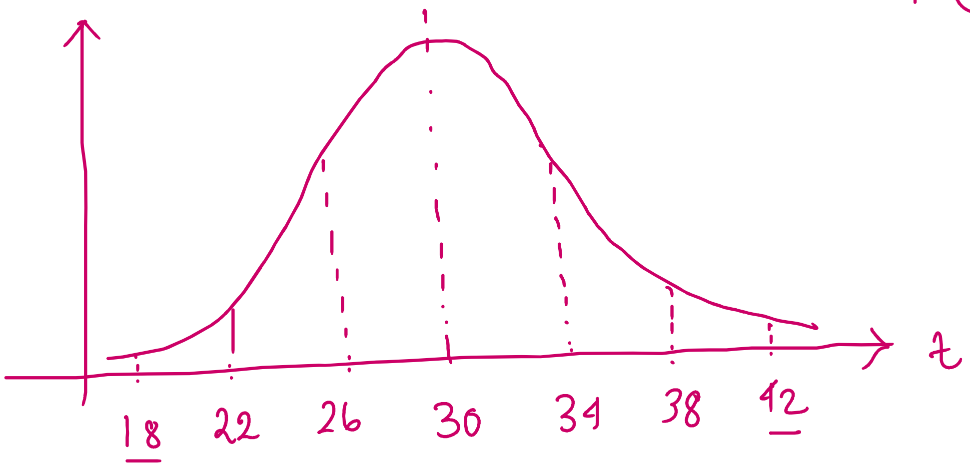
$$Z = \frac{x-\mu}{\sigma}$$

$$= \frac{100-50}{15} = \frac{50}{15} = 3.33 > 3$$

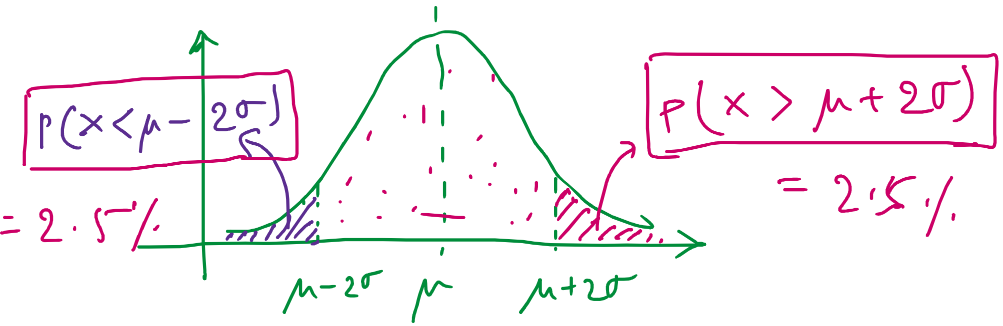
$T \sim N(\overset{\mu}{30}, \overset{\sigma^2}{16})$ $\quad \sigma = 4$

$\boxed{T_c} = ?$

$P(T \le T_c) = \underline{0.975}$

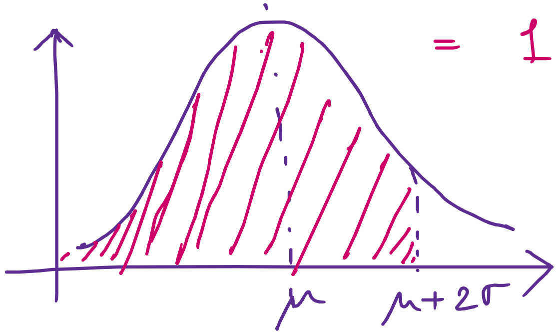


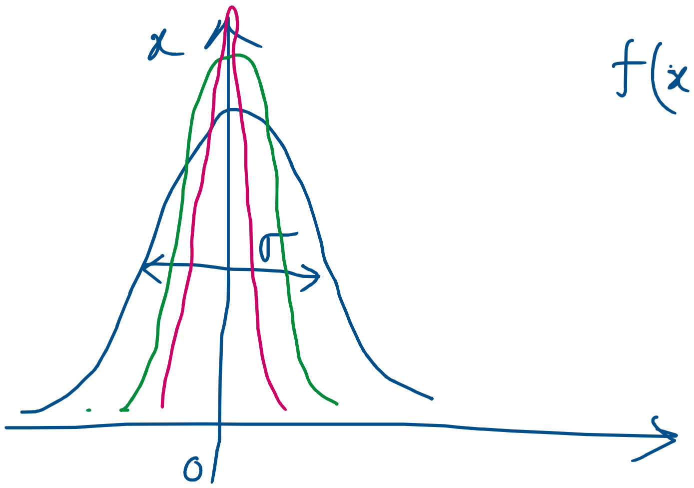
$T_c = 30 + 2 \times 4 = 38$



$P(X \le \mu + 2\sigma) = 1 - P(X > \mu + 2\sigma)$

$= 1 - 0.025 = \underline{0.975}$





$$f(x) = \frac{1}{\sqrt{2\pi}\,(\sigma)} e^{-x^2/2\sigma^2}$$

as $\sigma \to 0$. the $f(x) \to \delta(x)$

impulse function